

Supplementary Information

Systematic screening of brain neurotransmitter enzymes for radical pair mechanism competence identifies candidates for magnetic field sensitivity

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1 Derivation of key equations

1.1 Spin dephasing T_2^* from HFC distribution (Anderson–Weiss)

For an electron spin coupled to N nuclear spins with isotropic hyperfine coupling constants A_i and nuclear spin quantum numbers I_i , the inhomogeneous transverse relaxation rate in the Anderson–Weiss model is[?]:

$$\frac{1}{T_2^*} = \sqrt{\sum_{i=1}^N \frac{A_i^2 I_i(I_i + 1)}{3}} \quad (\text{S1})$$

where A_i is in angular frequency units (rad/s). This arises from the second moment of the HFC distribution:

$$M_2 = \sum_i A_i^2 I_i(I_i + 1)/3 \quad (\text{S2})$$

In the rigid-lattice (static) limit, $1/T_2^* = \sqrt{M_2}$. For MAO-A with $A(^{31}\text{P}) = 200$ MHz and two ^{31}P nuclei ($I = 1/2$):

$$\begin{aligned} M_2 &= 2 \times (200 \times 2\pi)^2 \times \frac{1}{2} \times \frac{3}{2} \times \frac{1}{3} \\ &= 2 \times (1.257 \times 10^9)^2 \times 0.25 = 7.90 \times 10^{17} \text{ rad}^2/\text{s}^2 \\ T_2^* &= 1/\sqrt{M_2} = 1.12 \text{ ns} \end{aligned} \quad (\text{S3})$$

1.2 HFC secular modulation T_2^A

The homogeneous contribution from secular (non-oscillatory) HFC modulation by molecular tumbling is[?]:

$$\frac{1}{T_2^A} = \frac{1}{12} \sum_i A_i^2 I_i(I_i + 1) \tau_c \quad (\text{S4})$$

where τ_c is the rotational correlation time. This term is *independent* of external magnetic field B_0 , unlike the g -anisotropy contribution. For MAO-A with $\tau_c = 10$ ns:

$$T_2^A = \left[\frac{1}{12} \times 2 \times (200 \times 2\pi \times 10^6)^2 \times 0.75 \times 10^{-8} \right]^{-1} = 0.50 \text{ ns} \quad (\text{S5})$$

1.3 g -anisotropy relaxation T_2^g

The transverse relaxation from modulation of the g -tensor anisotropy by molecular rotation is[?]:

$$\frac{1}{T_2^g} = \frac{1}{5} (\Delta g)^2 \omega_0^2 \tau_c \quad (\text{S6})$$

where $\Delta g = \xi_{\text{SOC}}/\Delta E$ is the g -shift from spin–orbit coupling and $\omega_0 = g\mu_B B_0/\hbar$ is the electron Larmor frequency. This term scales as $\omega_0^2 \propto B_0^2$, which is the key physics underlying the Earth-field advantage:

$$\frac{T_2^g(\text{Earth})}{T_2^g(\text{X-band})} = \left(\frac{B_X}{B_{\text{Earth}}} \right)^2 = \left(\frac{0.34 \text{ T}}{50 \mu\text{T}} \right)^2 = 4.6 \times 10^7 \quad (\text{S7})$$

At Earth’s field, $T_2^g \rightarrow \infty$ for all organic radicals, and this channel effectively vanishes.

1.4 Total electron T_2^e

The five relaxation channels combine as independent rates:

$$\frac{1}{T_2^e} = \frac{1}{T_2^*} + \frac{1}{T_2^{\text{dd}}} + \frac{1}{T_2^g} + \frac{1}{T_2^A} + \frac{1}{T_2^{\text{SOC}}} \quad (\text{S8})$$

1.5 Paramagnetic relaxation enhancement (PRE)

The Solomon–Bloembergen PRE for a nuclear spin at distance r from a paramagnetic electron is[? ?]:

$$\frac{1}{T_{1,\text{PRE}}} = \frac{2}{15} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma_I^2 g^2 \mu_B^2 S(S+1)}{r^6} \left[\frac{3\tau_c}{1 + \omega_I^2 \tau_c^2} + \frac{7\tau_c}{1 + \omega_S^2 \tau_c^2} \right] \quad (\text{S9})$$

$$\frac{1}{T_{2,\text{PRE}}} = \frac{1}{15} \left(\frac{\mu_0}{4\pi} \right)^2 \frac{\gamma_I^2 g^2 \mu_B^2 S(S+1)}{r^6} \left[4\tau_c + \frac{3\tau_c}{1 + \omega_I^2 \tau_c^2} + \frac{13\tau_c}{1 + \omega_S^2 \tau_c^2} \right] \quad (\text{S10})$$

The r^{-6} dependence creates the critical PLP/FAD dichotomy:

$$\frac{T_{2,\text{PRE}}(\text{FAD}, r = 7 \text{ \AA})}{T_{2,\text{PRE}}(\text{PLP}, r = 3.5 \text{ \AA})} = \left(\frac{7}{3.5} \right)^6 = 64 \quad (\text{S11})$$

1.6 Cross-relaxation rates (Solomon equation)

For a coupled e^- - ^{31}P spin pair, the Solomon cross-relaxation rates are[?]:

$$W_0 = \frac{1}{20} d^2 6J(\omega_S - \omega_I) \quad (\text{S12})$$

$$W_2 = \frac{3}{10} d^2 J(\omega_S + \omega_I) \quad (\text{S13})$$

where $d^2 = (\mu_0/4\pi)^2 \gamma_I^2 g^2 \mu_B^2 \hbar^2 / r^6$ and $J(\omega) = \tau_c / (1 + \omega^2 \tau_c^2)$ is the spectral density function. At Earth's field, $\omega_S \tau_c = 0.088 \ll 1$ (extreme narrowing), so $J(\omega_S) \approx \tau_c$, giving:

$$\frac{W_0(\text{Earth})}{W_0(\text{X-band})} \approx \frac{\tau_c}{[\tau_c / (1 + \omega_S^2 \tau_c^2)]_X} = 1 + \omega_{S,X}^2 \tau_c^2 \approx 3.5 \times 10^5 \quad (\text{S14})$$

1.7 Singlet yield (Green's function method)

For a radical pair with Hamiltonian $\hat{H}$, the time-integrated singlet yield with exponential recombination (rate k) is[?]:

$$\Phi_S = k \int_0^\infty \text{Tr}[\hat{P}_S \hat{\rho}(t)] e^{-kt} dt \quad (\text{S15})$$

where $\hat{P}_S = \frac{1}{4}\hat{1} - \mathbf{S}_1 \cdot \mathbf{S}_2$ is the singlet projection operator and $\hat{\rho}(t) = e^{-i\hat{H}t} \hat{\rho}_0 e^{i\hat{H}t}$. Diagonalising $\hat{H} = \hat{V} \hat{\Lambda} \hat{V}^\dagger$ with eigenvalues $\{E_m\}$:

$$\hat{\rho}(t) = \hat{V} e^{-i\hat{\Lambda}t} \hat{V}^\dagger \hat{\rho}_0 \hat{V} e^{i\hat{\Lambda}t} \hat{V}^\dagger \quad (\text{S16})$$

$$\Phi_S = k \sum_{m,n} [\tilde{P}_S]_{mn} [\tilde{\rho}_0]_{nm} \int_0^\infty e^{-i(E_m - E_n)t} e^{-kt} dt \quad (\text{S17})$$

$$= k \sum_{m,n} \frac{[\tilde{P}_S]_{mn} [\tilde{\rho}_0]_{nm}}{k + i(E_m - E_n)} \quad (\text{S18})$$

where $\tilde{O} = \hat{V}^\dagger \hat{O} \hat{V}$ denotes the operator in the energy eigenbasis. This is the expression used in Sim 1, avoiding the need for time-stepping propagation.

1.8 RPM quality metric Q_{RPM}

The composite quality metric combines all RPM-relevant parameters:

$$Q_{\text{RPM}} = f_{\text{osc}} \times T_2^e(\text{ns}) \times n_{\text{P}} \times \frac{T_2(^{31}\text{P})}{1 \mu\text{s}} \times \left(\frac{100 \text{ cm}^{-1}}{\xi_{\text{SOC}}} \right)^2 \quad (\text{S19})$$

The physical rationale for each factor:

- f_{osc} : proxy for electronic coupling strength at the radical site. For photochemical enzymes (CRY), this directly reflects RP generation probability; for dark enzymes (MAO-A), it serves as an approximate indicator of the cofactor’s capacity for electron transfer, though the correspondence is indirect
- T_2^e : time window for S-T mixing (spin evolution must occur before dephasing)
- n_P : number of HFC-active ^{31}P nuclei (S-T mixing drivers)
- $T_2(^{31}\text{P})$: nuclear spin coherence (magnetic field information is encoded here)
- $1/\xi^2$: penalty for strong SOC (destroys spin correlation via T_2 and ISC)

1.9 Coupled e^- - ^{31}P mixing angle

The zero-quantum transition dipole moment depends on the state mixing between $|\alpha_e\beta_n\rangle$ and $|\beta_e\alpha_n\rangle$. The mixing Hamiltonian in the $\{|\alpha_e\beta_n\rangle, |\beta_e\alpha_n\rangle\}$ subspace is:

$$H_{\text{ZQ}} = \begin{pmatrix} \omega_S/2 - \omega_I/2 & A/2 \\ A/2 & -\omega_S/2 + \omega_I/2 \end{pmatrix} \quad (\text{S20})$$

The mixing angle θ satisfies:

$$\tan(2\theta) = \frac{A}{\omega_S - \omega_I} \quad (\text{S21})$$

At Earth’s field, $A = 200 \text{ MHz} \gg \omega_S = 1.4 \text{ MHz}$, so $\tan(2\theta) \approx 143$ and $\theta \rightarrow \pi/4$. The forbidden ZQ transition dipole moment is:

$$\mu_{\text{ZQ}} = \sin(\theta) g\mu_B \sqrt{S(S+1)} = \frac{1}{\sqrt{2}} \times 1.734 \mu_B = 1.226 \mu_B \quad (\text{S22})$$

which is 70.7% of the allowed EPR TDM ($1.734 \mu_B$). This near-unity ratio is the defining feature of the strong-coupling regime and is essential for efficient ^{31}P -driven S-T interconversion at geomagnetic field strengths.

2 Quantum reservoir computing: loss function and readout

As a speculative extension—not a claim of this paper—we note that the multi-spin radical pair system could in principle function as a quantum reservoir for computation, analogous to proposals in condensed-matter spin systems. We include the following formalism for completeness; experimental validation is beyond the scope of this work.

The QRC readout uses Ridge regression with loss function:

$$\mathcal{L}(W) = \|XW - y\|_2^2 + \alpha \|W\|_2^2 \quad (\text{S23})$$

where $X \in \mathbb{R}^{N_{\text{train}} \times d}$ is the feature matrix (reservoir observables), $y \in \{0, 1\}^{N_{\text{train}}}$ is the label vector, $W \in \mathbb{R}^d$ is the weight vector, and $\alpha = 1.0$ is the regularisation parameter. The closed-form solution is:

$$W^* = (X^T X + \alpha I)^{-1} X^T y \quad (\text{S24})$$

Classification threshold: $\hat{y} = \mathbb{I}[XW > 0.5]$.

Table S1: Comparison of computed parameters with experimental values where available.

Parameter	System	This work	Experimental	Ref.
$A(^{31}\text{P})$	FAD semiquinone	200 MHz (input)	180–220 MHz	[?]
$T_2^*(^{31}\text{P})$	FADH $\bullet$	1.1 ns	~ 1 ns	[?]
$f_{\text{osc}}(S_0 \rightarrow S_1)$	FAD	0.50	0.23–0.30	[?]
$\lambda_{\text{abs}}(S_0 \rightarrow S_1)$	FAD	450 nm †	450 nm	[?]
ξ_{SOC}	Organic radical	63–79 cm $^{-1}$	40–80 cm $^{-1}$	[?]
T_1^e	Flavin radical	~ 100 μs	10–500 μs	[?]
MFE (flavoenzyme)	Various	–4.4% (peak)	1–15%	[?]

† Literature value used for Strickler–Berg calculation; xTB gap (1.25 eV $\approx$ 990 nm) underestimates.

3 Extended Data: Computed vs experimental values

4 Spin Hamiltonian for RPM simulation

The full spin Hamiltonian for the 8-spin radical pair simulation (Sim 1) is:

$$\hat{H} = \underbrace{J \mathbf{S}_1 \cdot \mathbf{S}_2}_{\text{exchange}} + \underbrace{\sum_{i \in \text{rad1}} A_i \mathbf{S}_1 \cdot \mathbf{I}_i}_{\text{HFC (radical 1)}} + \underbrace{\sum_{j \in \text{rad2}} A_j \mathbf{S}_2 \cdot \mathbf{I}_j}_{\text{HFC (radical 2)}} + \underbrace{g\mu_B B (\hat{S}_{1z} + \hat{S}_{2z})}_{\text{electron Zeeman}} + \underbrace{\sum_k \gamma_k B \hat{I}_{kz}}_{\text{nuclear Zeeman}} \quad (\text{S25})$$

Spin assignment for the 8-spin system:

Index	Spin	Type	A (MHz)	Assignment
0	S_1	electron	—	FADH $\bullet$ radical
1	S_2	electron	—	Substrate amine radical
2	I_1	^{31}P	200	FAD FMN-phosphate (proximal)
3	I_2	^{31}P	2	FAD AMP-phosphate (remote)
4	I_3	^1H	2.7	Isoalloxazine H
5	I_4	^1H	2.7	Isoalloxazine H
6	I_5	^1H	5.0	Substrate H
7	I_6	^1H	5.0	Substrate H

Hilbert space dimension: $2^8 = 256$. Exchange coupling: $J = 0$ (separated radical pair). Recombination rate: $k = 1$ MHz (spin-selective singlet recombination).

5 Thermodynamic energy budget of the three-layer model

A critical test of the three-layer quantum–classical hierarchy is whether the energy budget is self-consistent at each layer and across layer interfaces. We examine five specific questions below, identifying the energy sources, the role of magnetic fields, and the assumptions required.

5.1 Energy scales across the hierarchy

Table S2 summarises the energy scales relevant to each layer and interface. The hierarchy spans 14 orders of magnitude, from the ^{31}P nuclear Zeeman splitting (3.56 peV) to covalent bond energies (~ 4 eV).

Table S2: Energy scales in the three-layer hierarchy (310 K, $B = 50 \mu\text{T}$).

Quantity	Value	Layer	Role
^{31}P Zeeman splitting	3.56 peV (862 Hz)	L1	Nuclear spin state energy
^{31}P – ^{31}P dipolar	41 feV (0.01 MHz)	L1	Spin–spin coupling
HFC ($A_{31\text{P}}$)	$0.83 \mu\text{eV}$ (200 MHz)	L1 $\leftrightarrow$ L2	Nuclear–electron coupling
Electron Zeeman	5.79 neV (1.40 MHz)	L2	S– T_0 degeneracy lifting
Exchange coupling J	$0\text{--}0.41 \mu\text{eV}$ (0–100 MHz)	L2	S–T gap
SOC	7.81 meV (63 cm^{-1})	L2	ISC driving force
$k_B T$ (310 K)	26.7 meV	All	Thermal noise floor
5-HT oxidation	$\sim 0.8 \text{ eV}$	L2 $\rightarrow$ L3	Chemical energy source
MFE modulation	2.4 meV (0.3% of 0.8 eV)	L2 $\rightarrow$ L3	Redirected chemical energy
C–H bond	$\sim 4 \text{ eV}$	L3	Covalent bond scale

5.2 Q1: Radical pair generation energy

The radical pair (RP) is generated either by photoexcitation of FAD (2.76 eV, 450 nm) followed by electron transfer, or by single-electron transfer (SET) during substrate oxidation (~ 0.8 eV)[?]. Both pathways provide energy far exceeding $k_B T = 26.7$ meV, ensuring that RP formation is thermodynamically spontaneous. No external energy input beyond the substrate is required.

5.3 Q2: Energetics of singlet–triplet mixing

In the radical pair state with $J \approx 0$, the singlet $|S\rangle$ and triplet $|T_0\rangle$ states are nearly degenerate, separated by only $2J \lesssim 5.8$ neV at Earth’s field. The hyperfine-driven S– T_0 mixing is a *coherent quantum process* that transfers angular momentum between electron and nuclear spins without net energy exchange[? ?]:

$$|S\rangle \xrightleftharpoons{\text{HFC}} |T_0\rangle \quad (\text{iso-energetic, } \Delta E \lesssim 5.8 \text{ neV}) \quad (\text{S26})$$

This is analogous to Rabi oscillations between degenerate states driven by an internal coupling, and incurs no thermodynamic cost. The HFC energy ($0.83 \mu\text{eV}$) represents the coupling matrix element, not an energy that must be supplied externally.

5.4 Q3: Nuclear spin polarisation—the CIDNP mechanism

The thermal equilibrium polarisation of ^{31}P at Earth’s field is negligibly small:

$$P_{\text{eq}} = \tanh\left(\frac{gI\mu_N B}{2k_B T}\right) = 6.7 \times 10^{-11} \quad (\text{S27})$$

This would appear to preclude nuclear spins from carrying useful information. However, the radical pair mechanism generates *non-equilibrium* nuclear spin polarisation via CIDNP (Chemically Induced Dynamic Nuclear Polarisation)[? ?]. In CIDNP, the spin-selective recombination of radical pairs produces product

molecules whose ^{31}P nuclear spins are polarised far beyond the thermal equilibrium value. The CIDNP enhancement factor is:

$$\epsilon_{\text{CIDNP}} = \frac{P_{\text{CIDNP}}}{P_{\text{eq}}} \sim \frac{A}{g_I \mu_N B / \hbar} \sim \frac{200 \text{ MHz}}{862 \text{ Hz}} \sim 2.3 \times 10^5 \quad (\text{S28})$$

This $\sim 10^5$ -fold enhancement has been experimentally observed in flavin photochemistry[?]: photo-CIDNP of FMN produces ^{31}P polarisation enhancements of 10^3 – 10^5 over thermal equilibrium. The energy for this non-equilibrium polarisation comes from the chemical energy of the RP recombination reaction (~ 0.8 eV), not from the nuclear spin system itself. The polarisation decays with the nuclear T_1 (~ 4.6 s), providing a millisecond–second timescale quantum memory that is continuously refreshed by ongoing enzymatic turnover.

Assumption: MAO-A generates CIDNP during catalytic turnover. This is plausible because (a) MAO-A is a flavoprotein, (b) flavoproteins are known CIDNP sources[?], and (c) the SET mechanism in MAO involves radical pair intermediates[?]. However, direct observation of CIDNP in MAO-A has not been reported and represents a key experimental prediction.

5.5 Q4: Magnetic field as catalyst—no thermodynamic violation

The apparent “energy amplification” in the MFE is striking: a magnetic field perturbation of 5.8 neV (electron Zeeman at 50 μT) produces a 2.4 meV change in product energy partitioning, an amplification factor of $\sim 4 \times 10^5$. This does *not* violate the first or second law of thermodynamics because the magnetic field performs no work:

1. The Lorentz force $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ is perpendicular to the velocity; $\mathbf{F} \cdot \mathbf{v} = 0$.
2. The magnetic field shifts the S– T_0 degeneracy by $\Delta E = g\mu_B B$, changing the *rate* of S-T mixing but not the total energy of the system.
3. The chemical energy (0.8 eV per turnover) is provided entirely by 5-HT oxidation. The MFE changes how this energy is *partitioned* between singlet products (channel A) and triplet products (channel B), but the total energy released is:

$$\Delta G_{\text{total}} = \Phi_S(B) \cdot \Delta G_S + [1 - \Phi_S(B)] \cdot \Delta G_T \approx \text{const.} \quad (\text{S29})$$

provided $\Delta G_S \approx \Delta G_T$ (both channels release similar energy). Even when $\Delta G_S \neq \Delta G_T$, the energy difference is drawn from the substrate, not from the magnetic field. The magnetic field acts as a *catalyst*: it modifies the reaction pathway without contributing energy.

This is precisely the mechanism established by Steiner and Ulrich[?]: magnetic field effects on chemical reactions are kinetic (rate) effects, not thermodynamic (equilibrium) effects.

5.6 Q5: Information flow without energy flow

The three-layer cascade is thermodynamically consistent because each layer has its own energy source:

Layer transition	Information carrier	Energy source
L1 $\rightarrow$ L2	^{31}P spin state $\rightarrow$ S-T mixing rate	HFC (intrinsic molecular property)
L2 $\rightarrow$ L3	$\Phi_S(B) \rightarrow k_{\text{cat}}$ modulation	5-HT oxidation (0.8 eV/turnover)
L3 $\rightarrow$ neural	[5-HT] $\rightarrow$ receptor binding	ATP-driven synaptic machinery

The spin system functions as an *information channel*, not an energy channel. Information flows from nuclear spin states (Layer 1) through radical pair dynamics (Layer 2) to chemical concentrations (Layer 3), but the energy at each stage comes from local chemical sources. This is analogous to a transistor: a small gate voltage (information) controls a large drain current (energy from the power supply). In our system, the nuclear spin state (information) modulates the radical pair branching ratio (gate), which redirects chemical energy from 5-HT oxidation (power supply) between product channels.

5.7 Risk assessment of thermodynamic assumptions

Table S3: Risk assessment of energy budget assumptions.

Assumption	Risk level	Assessment
$J = 0$ (separated RP)	MODERATE	Valid for CRY Trp chain ($r > 10 \text{ \AA}$). Unknown for MAO-A active site ($r \sim 3\text{--}4 \text{ \AA}$). If $ J > A$ (200 MHz), MFE vanishes entirely. Testable by transient EPR.
$A(^{31}\text{P}) = 200 \text{ MHz}$	LOW	Experimental value for free flavin semiquinone. Protein environment may shift by $\pm 50\%$, but strong-coupling condition ($A \gg \omega_S$) holds for $A > 20 \text{ MHz}$.
$k = 1 \text{ MHz}$ (RP lifetime)	MODERATE	If RP recombines in $< 10 \text{ ns}$ ($k > 100 \text{ MHz}$), MFE decreases 100-fold. RP lifetime in MAO-A unknown; must be measured.
CIDNP generates polarisation	LOW	Well-established in flavoprotein photochemistry[?]. Expected but not directly observed in MAO-A catalysis.
B does no work	NONE	Fundamental physics: $\mathbf{F} \cdot \mathbf{v} = 0$ for magnetic Lorentz force[?].
Spin = information, not energy	NONE	Thermodynamically rigorous: spin selectivity partitions existing chemical energy.
0.3% rate change is detectable	HIGH	$\sim \text{nM}$ 5-HT concentration change over $\sim 10^{10}$ molecules. Whether this exceeds neural detection threshold is unknown.

In summary, the energy budget of the three-layer model is thermodynamically self-consistent. The magnetic field acts as a catalyst that redirects chemical energy between product channels via spin-selective radical pair chemistry; no energy is “created” or drawn from the field itself. The two moderate-risk assumptions ($J = 0$ and RP lifetime) and one high-risk assumption (neural detection threshold) are all experimentally testable and represent the critical unknowns for validating the model.